

2023 E2 CHIKOA

[B1.] $A+B = \begin{pmatrix} 2 & 8 & 2 & 9 \\ 2 & 6 & 2 & 11 \end{pmatrix}$

$$3A - 2B = \begin{pmatrix} 6 & -16 & 6 & -3 \\ -4 & 18 & -4 & 18 \end{pmatrix}$$

$$\begin{array}{l} 2(A+B) \\ 3A-2B \end{array} \Rightarrow \begin{array}{l} 2A+2B = 2(\quad) \\ 3A-2B = (\quad) \end{array}$$

$$5A \quad /$$

Beratz $5A = 2 \cdot (A+B) + 3A - 2B$

$$A = \frac{2}{5} (A+B) + \frac{3A-2B}{5}$$

$$5A = 2 \cdot \underbrace{\begin{pmatrix} 2 & 8 & 2 & 9 \\ 2 & 6 & 2 & 11 \end{pmatrix}}_{A+B} + \underbrace{\begin{pmatrix} 6 & -16 & 6 & -3 \\ -4 & 18 & -4 & 18 \end{pmatrix}}_{3A-2B}$$

$$5A = \begin{pmatrix} 10 & 0 & 10 & 15 \\ 0 & 30 & 0 & 40 \end{pmatrix}$$

$$A = \frac{1}{5} \begin{pmatrix} 10 & 0 & 10 & 15 \\ 0 & 30 & 0 & 40 \end{pmatrix} = \underline{\underline{\begin{pmatrix} 2 & 0 & 2 & 3 \\ 0 & 6 & 0 & 8 \end{pmatrix}}}$$

Beratz : $\rightarrow B = \begin{pmatrix} 2 & 8 & 2 & 9 \\ 2 & 6 & 2 & 11 \end{pmatrix} - A$

$$B = \begin{pmatrix} 2 & 8 & 2 & 9 \\ 2 & 6 & 2 & 11 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 2 & 3 \\ 0 & 6 & 0 & 8 \end{pmatrix} \Rightarrow$$

$$B = \underline{\underline{\begin{pmatrix} 0 & 8 & 0 & 6 \\ 2 & 0 & 2 & 3 \end{pmatrix}}}$$